QUASI-COMPACTNESS OF A REPRESENTABLE MORPHISM OF ALGEBRAIC SPACES TESTED AFTER BASE CHANGE FROM QUASI-COMPACT SPACES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is short but complete in both directions; the second direction uses the model case of an affine open of a scheme, which is exactly what the definition of quasi-compactness of a morphism of algebraic spaces asks for.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Quasi-compactness of an algebraic space tested by a surjective etale cover. Properties of Spaces, Definition spaces-properties-definition-quasi-compact, used at proof lines 1226-1230 and 1234-1238. Reduces quasi-compactness of the base change to a scheme cover.
- Quasi-compactness in the representable sense. Section section-representable, used at proof lines 1218-1219 and 1231-1237. Fixes the meaning of condition (1) and gives quasi-compactness of U x_Y X -> U.
- Quasi-compactness of a morphism of schemes tested on affine opens of the target. Schemes, Section schemes-section-quasi-compact, used at proof lines 1241-1246. Converts quasi-compactness of the preimages into quasi-compactness of p.

Source reference. Original section 01K2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2035–2035. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 03HA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4559–4559. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 03E3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4442–4442. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a representable morphism of algebraic spaces over S. The following are equivalent:*

(1) *f is quasi-compact (in the sense of Section 03HA), and*
(2) *for every quasi-compact algebraic space Z and any morphism $Z \to Y$ the algebraic space $Z \times_Y X$ is quasi-compact.*

Proof. Assume (1), and let $Z \to Y$ be a morphism of algebraic spaces with Z quasi-compact. By Properties of Spaces, Definition 03E3 there exists a quasi-compact scheme U and a surjective étale morphism $U \to Z$. Since f is representable and quasi-compact we see by definition that $U \times_Y X$ is a scheme, and that $U \times_Y X \to U$ is quasi-compact. Hence $U \times_Y X$ is a quasi-compact scheme. The morphism $U \times_Y X \to Z \times_Y X$ is étale and surjective (as the base change of the representable étale and surjective morphism $U \to Z$, see Section 03HA). Hence by definition $Z \times_Y X$ is quasi-compact.

Assume (2). Let $Z \to Y$ be a morphism, where Z is a scheme. We have to show that $p : Z \times_Y X \to Z$ is quasi-compact. Let $U \subset Z$ be affine open. Then $p^{-1}(U) = U \times_Y Z$ and the scheme $U \times_Y Z$ is quasi-compact by assumption (2). Hence p is quasi-compact, see Schemes, Section 01K2. □